Guide-field normalization and cascade growth in collisionless magnetic-island merging

Taylor Kalin

Independent researcher; https://taylorkalin.com

ORCID: 0009-0003-9000-9315; correspondence: teylersf@gmail.com

Version 1.0.0; 29 October 2026

Abstract

We examine magnetic-field normalization in an averaged model of collisionless magnetic-island coalescence. The target is the published study of Liu, Zhou and Loureiro and its earlier arXiv edition. Both estimate firehose crossing from a total-field ratio containing a uniform guide field, then identify its inverse with in-plane cascade gain. Their stated model instead requires a different conversion. We derive it, separate a missing factor of two in the printed temporal exponent, and retain the scalar firehose equation and guide-field accessibility criterion. The journal's added drive argument also applies a guide-independent temperature-ratio decrement to an in-plane merger; its own closure gives a guide-dependent factor and susceptibility. We state the local timescale consequences without claiming a universal replacement arrest criterion. Adding a uniform out-of-plane guide field also preserves the stated initial scalar-pressure magnetohydrodynamic force balance, although kinetic dynamics can change. Conditional numerical checks and independent high-precision calculations illustrate these identities. The journal's kinetic-firehose calculation, local diagnostics and model-applicability qualifications are credited. The corrections concern analytic growth, drive normalization and an equilibrium explanation; they do not provide corrected particle-in-cell spectra or disprove suppression of merging.

1. Introduction

Magnetic-island mergers can transfer magnetic energy toward larger scales while the reconnecting field decays. Analytical merger models connect these changes through the evolution of a characteristic in-plane field and island size [4,5]. A uniform guide field introduces another magnetic contribution: it influences magnetization and pressure anisotropy without necessarily sharing the decay of the reconnecting field. A scale-growth estimate must therefore specify which magnetic amplitude it uses.

Liu, Zhou and Loureiro investigate this problem with kinetic simulations and an approximate volume-averaged model [1,2]. Their published Eq. (C11) and Fig. 7(b) convert a total-field ratio into a cascade gain. We examine that conversion and the explanation of an early transient in the final paragraph of Appendix C [1, pp.9-10]. The earlier preprint contains the corresponding expressions as Eq. (B11) and the closing Appendix B paragraph [2].

Our question is conditional: what follows if the source's averaged closure and in-plane merger scaling are accepted? This separates an internally checkable algebraic correction from the harder problem of validating a kinetic closure or diagnosing a simulation. The analysis also treats the journal article as a substantive revision, rather than attributing omissions in the preprint to the published work.

2. Source editions and model assumptions

The primary source is the ten-page journal version of record, *The Astrophysical Journal Letters* **995**, L46 (2025), DOI 10.3847/2041-8213/ae247e [1]. We also inspect the twelve-page arXiv:2510.01573v1, submitted on 2 October 2025 [2]. The supplied preprint matches the official versioned PDF byte for byte. Edition hashes and the figure concordance are provided with the supporting materials.

The journal adds a distinction between fluid and kinetic firehose thresholds, an anisotropy-drive versus instability-growth comparison in Eqs. (2)-(6), and a pair-plasma dispersion calculation in Appendix B and Fig. 6 [1, pp.3-5,7-8]. Its new Fig. 4 provides local anisotropy diagnostics. The CGL discussion moves from preprint Appendix B to journal Appendix C; Fig. 7 adds an explicit region where insufficient magnetization limits that description. The preprint's separate well-magnetized, finite-guide appendix is absent from the journal edition. These changes matter when judging the evidence and limitations already supplied by the authors.

The source adopts double-adiabatic relations associated with the collisionless, heat-flux-neglected CGL description [3], then separately assumes approximate conservation for products of volume averages [1, Appendix C]:

$$\frac{\langle p_\perp \rangle}{\langle n \rangle \langle B \rangle} \approx \text{const.}, \qquad \frac{\langle p_\parallel \rangle \langle B \rangle^2}{\langle n \rangle^3} \approx \text{const.}$$

With approximately constant density and initially isotropic temperatures, its modeled total-field ratio $r = \mathcal{B}(t)/\mathcal{B}(t_0)$ gives

$$\frac{\langle T_\perp(t) \rangle}{T_0} \approx r, \qquad \frac{\langle T_\parallel(t) \rangle}{T_0} \approx r^{-2}, \qquad \langle \Delta \rangle \approx r^3 - 1, \qquad \langle \beta_\parallel \rangle \approx \beta_0 r^{-4}.$$

Here $\mathcal{B}$ denotes the total magnetic amplitude used by the source model. We do not identify an exact average of field magnitudes with a root-mean-square field. Factoring local quantities into separate averages is an explicit closure assumption, not a mathematical consequence of local adiabatic invariance.

For $u = 1 + (t - t_0)/\tau_0$, the source prescribes $B_{\rm in}(t) \approx (B_0/\sqrt{2})u^{-\alpha}$, with $\alpha \approx 0.42$, and estimates $\mathcal{B}^2(t) \approx B_{\rm in}^2(t) + B_G^2$ [1, Eq.(1), Appendix C]. The associated merger premise is $k(t)/k_0 \approx B_{\rm in}(t)/B_{\rm in}(t_0)$ [1,4,5]. We test the consistency of converting these modeled amplitudes, without claiming that the closure is quantitatively exact for the simulated plasma.

3. Total-field decay and in-plane scale growth

Let $b = B_G/B_0$, $s = B_{\rm in}(t)/B_{\rm in}(t_0)$ and $r = \mathcal{B}(t)/\mathcal{B}(t_0)$. The initial modeled in-plane amplitude is $B_0/\sqrt{2}$. Retaining the source's quadrature prescription therefore gives

$$r^2 = \frac{2b^2 + s^2}{1 + 2b^2}, \qquad s^2 = (1 + 2b^2)r^2 - 2b^2. \tag{1}$$

These are algebraic identities of that prescription, not identities for arbitrary volume averages. The source's merger scaling relates its characteristic wavenumber to the in-plane amplitude. With consistent initial and final wavenumber units, the required scale gain is

$$K \equiv \frac{k_0}{k(t)} \approx \frac{1}{s} = \left[(1 + 2b^2)r^2 - 2b^2\right]^{-1/2}. \tag{2}$$

Equation (C11) of the journal instead identifies K with $1/r$ [1,p.9]. The earlier Eq. (B11) makes the same identification [2,p.9]. For $b > 0$ and $0 < s < 1$,

$$r^2 - s^2 = \frac{2b^2(1-s^2)}{1+2b^2} > 0, \qquad \frac{1}{s} > \frac{1}{r}. \tag{3}$$

Thus the inverse total amplitude systematically understates the growth prescribed by the same in-plane merger model. It is exact in the zero-guide limit $b = 0$ and at the initial instant $s = 1$. A guide field can make the total magnitude nearly constant even while the reconnecting component decreases substantially. There is no general small relative error near the model's firehose-accessibility boundary.

Firehose accessibility remains a separate result

The averaged relations in Section 2 give the fluid firehose boundary as

$$f(r;\beta_0) \equiv r^3 + \frac{2}{\beta_0} r^4 - 1 = 0. \tag{4}$$

For every positive β_0, f is strictly increasing for $r > 0$, with $f(0) = -1$ and $f(1) > 0$. There is a unique root $r_* \in (0,1)$. The prescribed in-plane decay permits only

$$r > r_{\min} = \sqrt{\frac{2b^2}{1+2b^2}} \quad \text{at finite time.} \tag{5}$$

Consequently, $G = f(r_{\min};\beta_0) < 0$ gives a finite crossing in this model. Equality gives an asymptotic approach to marginality, and $G > 0$ makes the root inaccessible. This preserves the source's criterion while distinguishing finite crossing from equality at infinite modeled time. The conversion in Eq. (2) changes the growth associated with an accessible root, not Eq. (4) or the curve $G = 0$.

For fixed $b > 0$, write $q = 2b^2/(1+2b^2)$. The critical beta is

$$\beta_c(b) = \frac{2q^2}{1-q^{3/2}}. \tag{6}$$

As $\beta_0 \downarrow \beta_c$ from the finite-crossing side, $r_* \downarrow r_{\min}$. The inverse total ratio stays bounded by $1/r_{\min}$, but Eq. (2) grows without bound in the formal model. Since the root changes smoothly with β_0 and its derivative is positive at this boundary, K scales as $(\beta_0 - \beta_c)^{-1/2}$. This is a property of the algebraic model, not a prediction of physically infinite structures. A finite simulation domain, changes in the closure and a different kinetic history can intervene first.

At fixed b and large β_0, both gains approach one. Expanding Eq. (4) gives $1 - r_* \sim 2/(3\beta_0)$; the leading small excess $K - 1$ is larger than $1/r_* - 1$ by the factor $1 + 2b^2$. A correction to a relative increment can therefore coexist with small absolute growth at very high beta. It does not overturn the source's broad high-beta motivation by itself.

A separate printed time-exponent inconsistency

The prescribed in-plane decay is $s = u^{-\alpha}$. Accordingly,

$$\frac{1}{r} = \left[\frac{1+2b^2}{2b^2+u^{-2\alpha}}\right]^{1/2}, \qquad K = u^{\alpha}. \tag{7}$$

Both printed B11 and C11 instead place $u^{-\alpha}$ in the denominator of the first expression. Their own B7/B8 or C7/C8 require the squared-field exponent -2α. Fixing that exponent restores the total-field time expression; it does not repair the identification of $1/r$ with K. Nor does a printed inconsistency establish which formula produced the published figure: the original plotting program and its execution were not obtained.

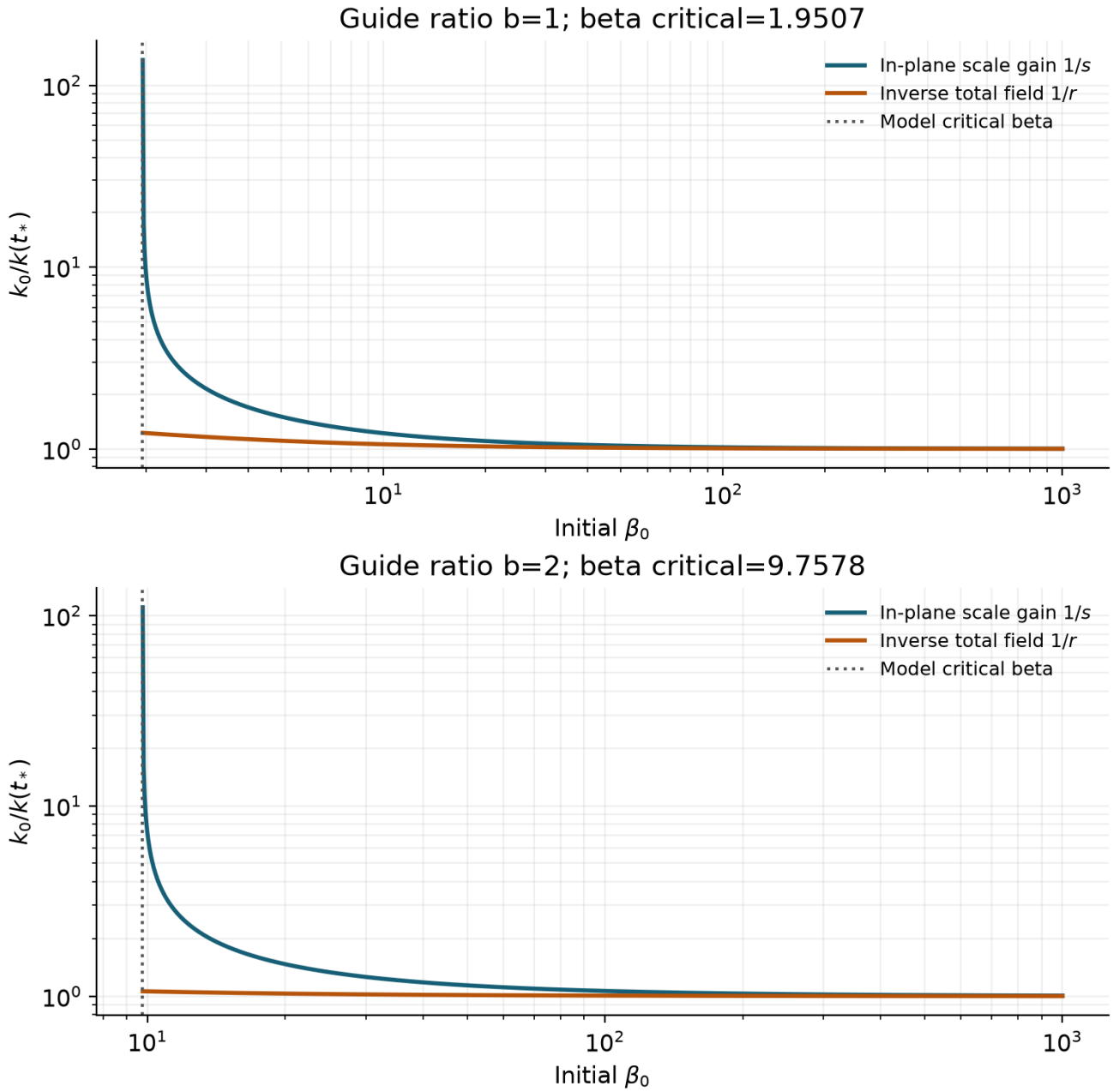


Figure 1. Model scale conversion for $b = 1$ and $b = 2$. The source's guide-field accessibility boundary remains unchanged. The divergent corrected curve is limited physically by finite size and model applicability, neither of which is imposed as an unmeasured numerical cutoff here.

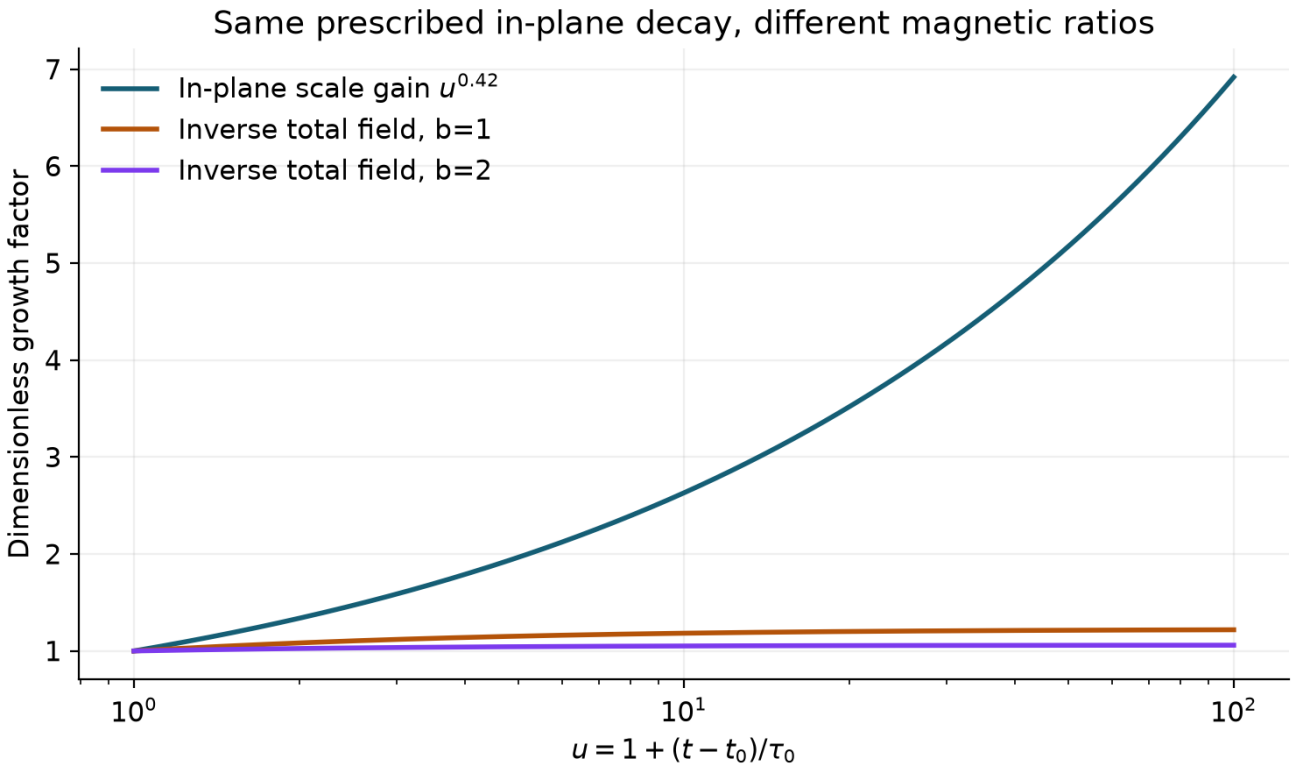


Figure 2. Ratio conversion for $\alpha = 0.42$. The horizontal coordinate is the source's normalized time u; no physical crossing time or post-arrest evolution is fitted.

4. Conditional numerical checks

The supplied reproduction solves Eq. (4) by monotone bisection and independently checks its positive root using the full quartic polynomial. It then evaluates the accessible root using Eq. (2). Thirteen crossing cases include the zero-guide limit, both finite guide ratios, inaccessible crossings, and successively closer approaches to each critical beta. Separate temporal checks include the initial $u = 1$ control. An inaccessible mathematical root is recorded with null physical-branch gain and time rather than plotted as an allowed crossing.

Guide ratio b	Initial β_0	Inverse total ratio $1/r_*$	In-plane gain K	Model crossing
0	10.8	1.055368	1.055368	Finite
1	3	1.163042	2.142550	Finite
1	2	1.220744	8.726781	Finite
2	10	1.059341	7.082716	Finite
2	100	1.006580	1.064357	Finite
2	1	1.353210	Not accessible	None

Table 1. Values of the stated analytic model, not corrected measurements of simulation spectra. The zero-guide row is a formal algebraic control outside the source's acknowledged CGL applicability near magnetic nulls. The $b = 1$, $\beta_0 = 3$ row uses the source's approximate initial parameters as an illustration; its transient and imperfect averaged closure still matter. The $b = 2$, $\beta_0 = 1$ case demonstrates why solving the scalar polynomial is insufficient without checking the accessible field interval.

For $b = 1$, $\beta_c = 1.950734$; for $b = 2$, $\beta_c = 9.757772$. The $b = 2$, $\beta_0 = 10$ example shows that a large conversion difference is not confined to a low-beta algebraic limit. It remains a conditional

closure calculation. It is not evidence that the physical system actually reaches that growth, that its instability remains unregulated, or that the observed spectral peak should be replaced by the table entry.

An independent calculation uses 80-digit decimal arithmetic and a separate equation in s^2. Its total-field roots agree to about 1.2×10^{-16} absolute. The closest critical cases amplify floating-point cancellation in the converted gain, with a maximum relative difference about 1.2×10^{-9}. The six-decimal table therefore avoids implying precision from the root residual alone. Machine-readable outputs retain the unrounded values and the comparison receipt.

The force-balance check in Section 5 is also exercised on a periodic 128^2 grid using Fourier derivatives, with source-normalized $B_0 = 0.5$, $T_0 = 0.25$, pair pressure $2nT_0$ and four constant guide amplitudes. Its maximum component residual is 1.40×10^{-14} or smaller. A deliberately nonuniform added guide produces a residual of 0.05. A separate analytic trigonometric evaluation confirms both the unchanged constant-guide force balance and the nonuniform negative control. This is a check of an initial field identity; grid size here does not represent a new kinetic simulation.

All checks run locally without fetching source PDFs or executing unknown source code. No particles are evolved, no dispersion solver is rerun and no statistical confidence interval is attached to these deterministic identities. The original source analysis scripts, simulation histories and plotting chain have not been recovered through this calculation. The illustrative clock uses $\alpha = 0.42$; the source reports different fitted slopes in its guide cases. No guide-specific time history is refitted here, and the root-to-scale algebra itself is independent of α.

Finite guide field in the journal's anisotropy-drive estimate

The journal adds a kinetic-firehose growth calculation and a comparison between instability growth and anisotropy drive [1,Section4,pp.4-5]. The idea of comparing these actual times is sound. Its intermediate assertion that every in-plane merger decreases $Q = T_\perp/T_\parallel$ by $2^{-3/2}$, however, drops the constant guide contribution. Define the modeled characteristic in-plane magnetic fraction

$$\chi = \frac{B_{\rm in}^2}{B_{\rm in}^2 + B_G^2}.$$

The same approximately fixed-density CGL closure gives $Q \propto \mathcal{B}^3$. For one prescribed merger $B_{\rm in} \mapsto B_{\rm in}/\sqrt{2}$,

$$\frac{Q_{\rm next}}{Q} = \left(\frac{B_G^2 + B_{\rm in}^2/2}{B_G^2 + B_{\rm in}^2}\right)^{3/2} = \left(1 - \frac{\chi}{2}\right)^{3/2}.$$

The source's factor follows only when $\chi = 1$. At its initial modeled amplitude, $\chi_0 = 1/(1 + 2b^2)$:

b	χ_0	Guide-dependent $Q_{\rm next}/Q$	Printed factor
0	1	0.353553	0.353553
1	1/3	0.760726	0.353553
2	1/9	0.917835	0.353553

Table 2. A single modeled merger, holding the guide field constant. This is a conditional temperature-ratio factor, not a fitted simulation measurement. As the in-plane field decays, χ also changes, so successive identical in-plane reductions do not produce a constant logarithmic decrement in Q.

The corresponding differential relation is

$$\frac{d\ln(1+\Delta)}{dt} = 3\chi\frac{d\ln B_{\rm in}}{dt} = -\frac{3\alpha\chi}{\tau_0 u}.$$

With $\tau_B = -(d\ln B_{\rm in}/dt)^{-1}$, a short interval near isotropy and large parallel beta gives the local estimate

$$t_{\rm cross} \simeq \frac{2\tau_B}{3\beta_\parallel \chi}.$$

This approximation requires that beta, χ and drive change little over the crossing interval; initially it needs $\beta_0\chi_0 \gg 1$. It is not applicable to an arbitrarily delayed crossing near $G = 0$ and does not replace the accessibility test. Under a constant fractional in-plane decay, identifying a merger time with a factor-$\sqrt{2}$ reduction changes an order-one coefficient. It cannot remove a parametrically small χ from the drive.

A related normalization needs care in the reduction from journal Eq.(5) to Eq.(6). The reconnection Alfvén speed is explicitly based on the in-plane field, whereas beta, gyrofrequency and Larmor radius use the total field. With consistent thermal and mass-density conventions, the scaling is $v_{\rm th}/v_{A,\rm in} \propto \sqrt{\beta/\chi}$, rather than $\sqrt{\beta}$. Combining this distinction with the local drive factor yields a conditional growth-to-drive parameter proportional to $(R/\rho)/(\beta\chi^{3/2})$, up to order-one constants and the assumed reconnection rate. It does not justify a universal replacement for Eq.(6): the actual distribution, growth rate and integrated drive would still need evaluation. The reproduction separately checks the exact generation factors and power-law derivative, leaving the original dispersion solver and physical arrest untouched.

5. Uniform-guide scalar-MHD force balance

The journal attributes an early transient to the initial configuration ceasing to be an exact MHD equilibrium when a guide field is included [1, p.10]. That explanation conflicts with the stated field geometry. This is a force-balance question independent of the averaged CGL approximation.

Write the initial field as $\mathbf{B}_{\rm in}(x,y) + B_G\hat{\mathbf{z}}$, where B_G is spatially uniform and the planar field has no z dependence. Ampère's law then gives

$$\mathbf{J} = \frac{c}{4\pi}\nabla\times\mathbf{B}_{\rm in} = J_z(x,y)\hat{\mathbf{z}}.$$

The added guide produces no current. Moreover,

$$\frac{\mathbf{J}\times(\mathbf{B}_{\rm in}+B_G\hat{\mathbf{z}})}{c} = \frac{\mathbf{J}\times\mathbf{B}_{\rm in}}{c},$$

because parallel vectors have zero cross product. Therefore, if the source's scalar pressure satisfies $\nabla p = \mathbf{J}\times\mathbf{B}_{\rm in}/c$ initially, the same pressure satisfies force balance after adding any constant B_G. This argument preserves whatever force residual the planar configuration already has; it does not require stability.

The explicit source profile confirms the result. With $q = 2\pi k_0/L$, its fields are $B_x = -B_0\cos(qx)\sin(qy)$ and $B_y = B_0\sin(qx)\cos(qy)$ [1, Section 2]. Their curl is solely out of plane. Equivalently, a constant guide adds spatially constant magnetic pressure and contributes no $(\mathbf{B}\cdot\nabla)\mathbf{B}$ term in this two-dimensional geometry.

This correction does not eliminate the reported transient. Scalar-pressure MHD balance is weaker than a stationary Vlasov distribution: it does not specify particle orbits, pressure-tensor evolution, discretization effects or relaxation. A guide field can alter those processes while leaving the initial scalar-MHD force unchanged. Identifying the actual cause would require kinetic initial-condition and simulation diagnostics beyond the present calculation. The supported conclusion is that the uniform guide alone cannot break the stated scalar-MHD balance.

6. Discussion

The main distinction is between the magnetic quantity that governs a modeled instability and the quantity that defines a merger scale. The guide field belongs in the total amplitude used by the source's anisotropy closure. It cannot be left inside a field-to-size conversion derived for the decaying reconnecting component. Equations (1)-(7) make that separation explicit while retaining the source's scalar firehose equation. The same distinction gives the finite-guide generation factor and susceptibility in Section 4; dropping it also makes the simplified anisotropy-drive estimate guide independent.

The published Fig. 7(b) is labeled as an inverse cascade ratio and is discussed as evidence that growth remains small over a wide range of beta [1]. The total-field proxy provides an artificial finite plateau as the in-plane field declines. The corrected model gain has a different behavior near its accessibility boundary. An analytic plot or quantitative statement about merger-scale gain should therefore use Eq. (2), identify its valid branch and explain the finite-box and closure limits. A total-field amplification plot can still be useful if labeled as that quantity.

This correction does not establish that the source's spectra or observed arrest are wrong. The final article supplies local pressure-anisotropy diagnostics, a kinetic growth calculation, and explicit applicability caveats. Its kinetic-firehose competition is a stronger argument than the preprint's earlier discussion, and the present deterministic checks do not replace that calculation. The justified correction is to an internal conversion and an equilibrium explanation, with any measured change to the physical growth left for a matched simulation reanalysis.

7. Limitations, reproducibility and disclosures

This audit is edition-specific. The supplied arXiv PDF and the authentic journal version were read in full, with equations and decisive paragraphs checked visually. Hashes, page locators and a substantive concordance identify them. One provisional size-ratio observation in the archived working concordance is excluded from the manuscript's findings because geometric conventions remain unresolved; `SOURCE_SCOPE.md` documents that boundary. The comparison does not claim a character-by-character editorial diff. Bounded correction searches found no identified erratum; they do not prove that no correction or inaccessible data deposit exists.

The main results are conditional on the source's own characteristic-field prescription, averaged CGL closure and in-plane merger scaling. The local kinetic invariants do not automatically imply the chosen separate-average relations. Density changes, heat flux, nonadiabatic motion, spatial intermittency and unstable fluctuations may affect their accuracy. Near a critical curve, a formally large model gain can reach the box scale or leave the closure's useful domain. No physical divergence is asserted.

A uniform guide's preservation of scalar-MHD force balance does not establish an exact Vlasov equilibrium, stability, or the absence of a genuine startup transient. It identifies what the stated scalar-MHD explanation cannot supply. Diagnosing the actual transient requires additional particle distributions, initialization details or matched numerical controls. The proof does not assign an undocumented numerical cause.

Code and data availability. The original analytic checks, deterministic JSON/CSV outputs, standalone independent verifiers, original figures and source-edition records are archived as software version 1.0.0 [6], `https://doi.org/10.5281/zenodo.23779360` , direct record `https://zenodo.org/records/23779360` . The frozen computational commit is `8ecf2c6407731de91025208153a269a59573e515` in `https://github.com/Teylersf/inverse-transfer-guide-field-audit` . Its 25-file ZIP contains 242,467 bytes, SHA-256 `f0f4396b3f2d534683d16a9c42d373071c86943a9d0f6e086eff7ccc0c645214`. A clean Git-tree extraction reproduced all four programs and preserved all 25 file bytes in the recorded environment. Exact byte identity is not promised for arbitrary platforms, library versions or fonts; the independent numerical tolerances remain material.

On 29 October 2026, the anonymous Zenodo API and archive download returned HTTP 200 with the expected creator, ORCID, software type, version, commit and component licenses. The primary DataCite record, `https://api.datacite.org/dois/10.5281/zenodo.23779360` , returned HTTP 200 and state `findable`; doi.org resolved to the public Zenodo record and its registration-authority service confirmed DataCite. `submissions/zenodo-code-v1.0.0.json` retains timestamped metadata and exact archive verification. The software DOI is distinct from paper acceptance and need not appear in Crossref to be valid.

Source PDFs remain in a separate local cache, linked by their official references rather than bundled as the author's work. Reproduction needs no original HPC path, private data, GPU or new PIC run. Venue submission, moderation, review and publication states are recorded separately from these computational checks.

Taylor Kalin is the canonical author and self-funds this study. No outside funding or competing interests have been reported. AI tools assisted with retrieval, derivations, code, adversarial computational checks and drafting. Those checks are not independent human peer review. Any separately authorized agent-only edition identifies its agent author and preserves Taylor Kalin's credit as human operator and canonical author. The source authors and their institutions have not endorsed this audit. Original audit code is MIT licensed; original text and figures are CC BY 4.0. Attributed source material retains its own rights.

8. Conclusion

A finite uniform guide field separates total-field decay from the decay of the reconnecting component. Within the source's stated model, the in-plane scale gain is $[(1+2b^2)r_*^2-2b^2]^{-1/2}$, rather than $1/r_*$. The difference is quantitative and can become large near the model boundary, although very-high-beta gains still approach one at fixed guide ratio. The corrected conversion retains the scalar firehose criterion and its accessible branch. The same field distinction supplies a guide-dependent anisotropy generation factor and drive susceptibility; its local timescale consequence is conditional and does not provide a universal replacement arrest criterion. A missing factor of two in the printed temporal exponent and an incorrect scalar-MHD explanation of the startup transient are separate, localized corrections. None of these identities alone supplies a corrected kinetic spectrum or overturns the observed suppression mechanism.

References

[1] Z. Liu, M. Zhou and N. F. Loureiro, *Suppression of Inverse Magnetic Energy Transfer in Collisionless Marginally Magnetized Plasmas*, The Astrophysical Journal Letters **995**, L46 (2025). https://doi.org/10.3847/2041-8213/ae247e . Version of record, online 11 December 2025; issue date 20 December 2025.

[2] Z. Liu, M. Zhou and N. F. Loureiro, *Suppression of inverse magnetic energy transfer in collisionless marginally magnetized plasmas*, arXiv:2510.01573v1 (2 October 2025). https://arxiv.org/abs/2510.01573v1 ; https://arxiv.org/pdf/2510.01573v1 . Earlier edition, identified separately from [1].

[3] G. F. Chew, M. L. Goldberger and F. E. Low, *The Boltzmann equation and the one-fluid hydromagnetic equations in the absence of particle collisions*, Proceedings of the Royal Society of London A **236**, 112-118 (1956). https://doi.org/10.1098/rspa.1956.0116 . Bibliographic metadata verified directly; the closure as used here is explicitly stated in [1].

[4] M. Zhou, P. Bhat, N. F. Loureiro and D. A. Uzdensky, *Magnetic island merger as a mechanism for inverse magnetic energy transfer*, Physical Review Research **1**, 012004(R) (2019). https://doi.org/10.1103/PhysRevResearch.1.012004 .

[5] M. Zhou, N. F. Loureiro and D. A. Uzdensky, *Multi-scale dynamics of magnetic flux tubes and inverse magnetic energy transfer*, Journal of Plasma Physics **86**, 535860401 (2020). https://doi.org/10.1017/S0022377820000641 .

[6] T. Kalin, *Computational evidence for Guide-field normalization and cascade growth in collisionless magnetic-island merging*, version 1.0.0, Zenodo (2026), software and data. https://doi.org/10.5281/zenodo.23779360 . Computational snapshot 8ecf2c6407731de91025208153a269a59573e515.